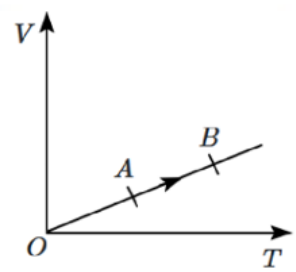
NEET Sample Paper (Model 10)

PHYSICS SECTION

1. The radius of the orbit corresponding to the 4^{th}excited state in Li^{++}ion is:
(where $a_0$is the radius of the first orbit in the hydrogen atom)

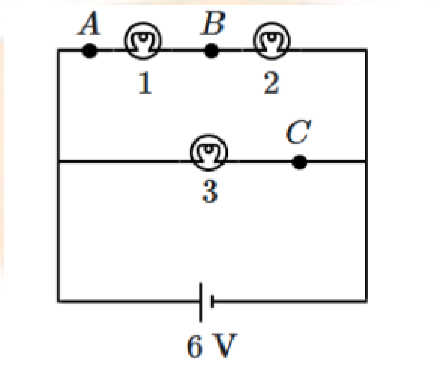
1. $\frac{25}{3}a_0$
2. $\frac{16}{3}a_0$
3. $25a_0$
4. $12a_0$

2. The volume $(V)$of a monoatomic gas varies with its temperature (T), as shown in the graph. The ratio of work done by the gas to the heat absorbed by it when it undergoes a change from state $A$to state Bwill be:

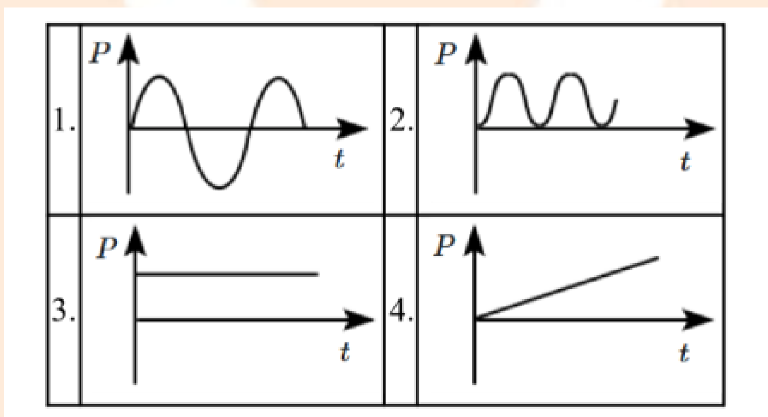


1.	$\frac{2}{5}$	2.	$\frac{2}{3}$
3.	$\frac{1}{3}$	4.	$\frac{2}{7}$

3. In the circuit shown, each light bulb has a resistance **2 Ω**, and the potential source has an **EMF of 6 V**.



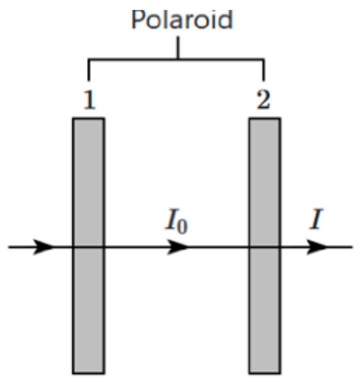
Which graph best shows the **power dissipated by the light bulb (3)** as a function of time?



4. Two identical polaroid sheets are held perpendicular to an unpolarised light beam. If I_0 is the intensity of the light between the two polaroids and I is the intensity of the emergent beam (see figure), then the intensity ratio,

$$\frac{I}{I_0} = \frac{1}{2}.$$

The angle between the pass axes of the two polaroids is:



1.	90°	2.	60°
3.	45°	4.	30°

5. Alternating equal and opposite charges, each of magnitude q, are placed at the six vertices of a regular hexagon of side a. If one of the negative charges is now removed, the electrostatic potential at the centre of the hexagon is:

$$\left(k = \frac{1}{4\pi\varepsilon_0}\right)$$

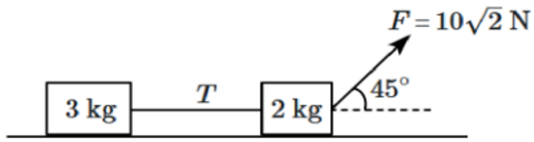
1. $k\frac{q}{a}$

2. $k\frac{2q}{a}$

3. $k\frac{\sqrt{3}q}{2a}$

4. $k\frac{q}{2a}$

6. The system of blocks, connected by an ideal horizontal string, lies at rest on a smooth horizontal surface. A constant force $F(= 10\sqrt{2}\text{ N})$acts on the 2 kg block, at an angle of $45°$above the horizontal. The system undergoes a displacement of 2m. Match the quantities mentioned in **Column I** with their values (SI) in **Column II**. Take $g = 10\text{ m/s}^2$.



	Column I		Column II
(A)	Work done by force F on 2 kg block	(I)	$20\sqrt{2}$
(B)	Work done by tension (T) on 2 kg block	(II)	12
(C)	Power due to force F, finally	(III)	20
(D)	Final kinetic energy of 3 kg block	(IV)	−12

1.	A-II, B-II, C-III, D-I
2.	A-III, B-IV, C-I, D-II
3.	A-I, B-IV, C-III, D-II
4.	A-II, B-IV, C-I, D-III

7. Two waves can be expressed as

$x = 4\sin\left(\omega t + kx + \frac{\pi}{3}\right)$and $x' = 3\cos\left(\omega t + kx\right)$.

The phase difference between the two waves is:

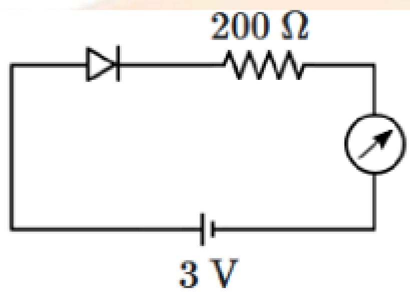
1. $\frac{\pi}{3}$rad
2. $\frac{\pi}{6}$rad
3. $\frac{\pi}{2}$rad
4. πrad

8. A 100-turn coil of wire of size 2 cm × 1.5 cmis suspended between the poles of a magnet producing a field of 1 T, inside a galvanometer. Calculate the torque on the coil due to a current of 0.1 Apassing through the coil.

1. 3×10^{-5}N·m
2. 30N·m
3. 3×10^{-3}N·m

4. 3×10^{-2}N·m

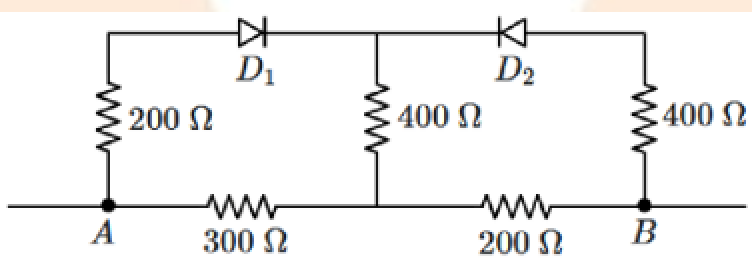
9. The ammeter reading for the silicon diode in the given circuit is:



1.	0	2.	15 mA
3.	11.5 mA	4.	13.5 mA

10. In the circuit shown in the figure, the diodes D_1, D_2 are ideal.

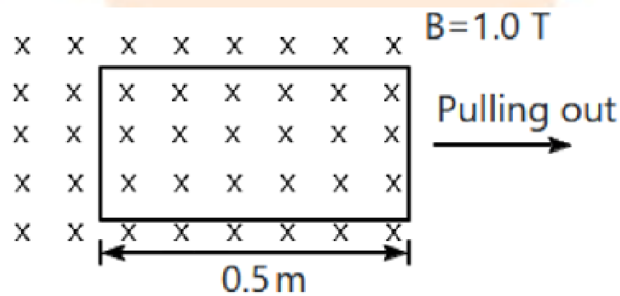
The resistance between A& B, when $V_A < V_B$, is:



1. 460 Ω
2. 400 Ω
3. 600 Ω
4. 1500 Ω

11. Eddy currents can be minimised by:

1. reducing relative motion between the core and magnet in an electric motor
2. by making the core of thin laminations
3. by increasing the conductor cross-sectional area
4. both (1) and (2) are correct



1.	3.125×10^{-3} J	2.	6.25×10^{-4} J
3.	1.25×10^{-2} J	4.	5.0×10^{-4} J

12. Select the correct statements regarding potential energy (U)in the simple harmonic motion of a particle along the x-axis:

1. $\frac{dU}{dx} < 0$for all positions of a particle performing SHM.
2. $\frac{dU}{dx} > 0$for all time.
3. Potential energy is minimum at the equilibrium position of a particle performing SHM.
4. Potential energy increases linearly with the position as the particle moves away from the equilibrium position.

13. A soap bubble of radius $r$is blown up to form a bubble of radius $2r$. If $\sigma$is the surface tension of soap solution, the energy spent in doing so is: (Assuming surface tension is constant.)

1. $3\pi\sigma r^2$
2. $6\pi\sigma r^2$
3. $12\pi\sigma r^2$
4. $24\pi\sigma r^2$

14. Figure shows a square loop of side **0.5 m** and resistance **10 Ω**. The magnetic field has a magnitude $B = 1.0\,T$. The work done in pulling the loop out of the field uniformly in **2.0 s** is:

1. 3.125×10^{-3}J
2. 6.25×10^{-4}J
3. 1.25×10^{-2}J
4. 5.0×10^{-4}J

15. A tuning fork, placed in a room, vibrates according to the equation:

$$Y = (10^{-4}\text{ m})\sin\left(\frac{2\pi t}{0.01\text{ s}}\right)$$

where $Y$is the displacement of the tip of a prong. The speed of sound in air is 330 m/s. The frequency of the tuning fork is:

1. 100 Hz
2. 50 Hz
3. 200 Hz

4. 200π Hz

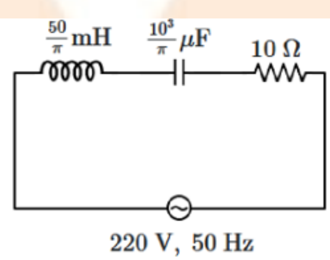
16. A mixture of two or more ideal gases:

1. does not follow the ideal gas equation
2. does not follow the first law of thermodynamics
3. has the same RMS speed (per molecule) for each gas
4. has the same average translational kinetic energy for each type of gas molecule

17. An electron falls from rest through a vertical distance $h$in a uniform and vertically upward-directed electric field E. The direction of the electric field is now reversed keeping its magnitude the same. A proton is allowed to fall from rest through the same vertical distance h. The fall time of the electron in comparison to the fall time of the proton is:

1. smaller
2. 5 times greater
3. 10 times greater
4. equal

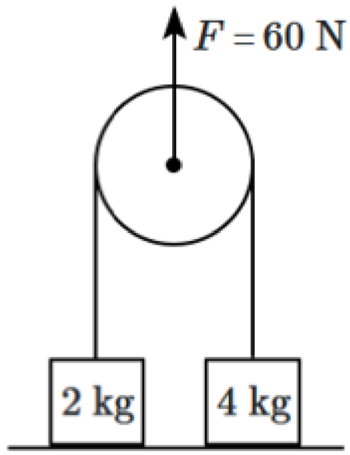
18. The net impedance of the circuit (as shown in the figure) will be:



1.	25 Ω	2.	$10\sqrt{2}$ Ω
3.	15 Ω	4.	$5\sqrt{5}$ Ω

19. A force of **60 N** is applied vertically to the system, where the pulley and string are ideal. The blocks are on the ground. The upward acceleration of the pulley is:

(take $g = 10\ \text{m/s}^2$)



1. $5\ \text{m/s}^2$
2. $10\ \text{m/s}^2$
3. $2.5\ \text{m/s}^2$
4. $1.5\ \text{m/s}^2$

20. In a gravitational field, the gravitational potential is given by; $V = -Kx$ J/kg. The gravitational field intensity at the point $(2, 0, 3)$m is:

1. $+\frac{K}{2}$
2. $-\frac{K}{2}$
3. $-\frac{K}{4}$
4. $+\frac{K}{4}$

21. In Young's double-slit experiment, if there is no initial phase difference between the light from the two slits, a point on the screen corresponding to the fifth minimum has a path difference:

1. $\frac{5\lambda}{2}$
2. $\frac{10\lambda}{2}$
3. $\frac{9\lambda}{2}$
4. $\frac{11\lambda}{2}$

22. Four identical point charges q, each are fixed at the corners of a square of side a. The work done in bringing a point charge q, from infinity and placing it at the centre of the square is: $\left(k=\frac{1}{4\pi\varepsilon_0}\right)$

1. $k\,\frac{2q^2}{a}$
2. $k\,\frac{4\sqrt{2}q^2}{a}$
3. $k\,\frac{4q^2}{a\sqrt{2}}$
4. zero

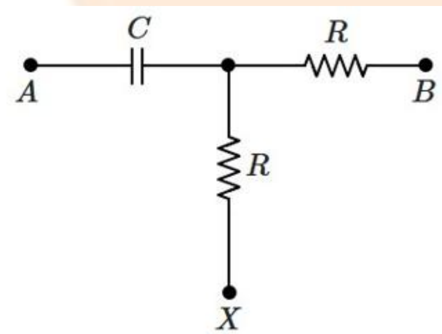
23. The de-Broglie wavelength of an electron in the ground state of a H-atom is λ_1 and that on the He $^+$ ion is λ_2. The kinetic energies of the electrons in the H-atom and the He $^+$ ion having the same de-Broglie wavelength λ_1 are E_1 and E_2. The ratio $\lambda_2 : \lambda_1$ is:

1. 4
2. 2

3. $\frac{1}{2}$

4. $\frac{1}{4}$

24. The end ‘B’ of the circuit is earthed ($V_B = 0$)while a sinusoidal voltage is applied at ‘A’, $V_A = V_0 \sin\ \omega t$. The rms voltage across the capacitor Cequals that across the upper resistor R(as shown in the figure). What is the phase difference between the current through the capacitor and the voltage across the capacitor when no current flows out at X?



1. 0°

2. 45°

3. 90°

4. 180°

25. Match List I with List II.

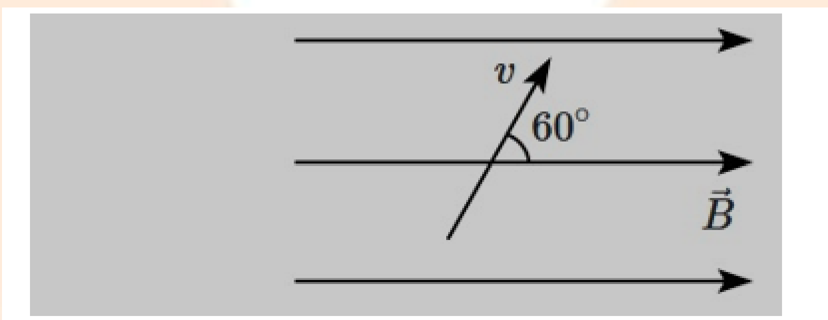
List I (Spectral Lines of Hydrogen for transitions from)
A. $n_2 = 3$ to $n_1 = 2$
B. $n_2 = 4$ to $n_1 = 2$
C. $n_2 = 5$ to $n_1 = 2$
D. $n_2 = 6$ to $n_1 = 2$

List II (Wavelength (nm))
I. 410.2
II. 434.1
III. 656.3
IV. 486.1

Choose the correct answer from the options given below:

1. A – III, B – IV, C – II, D – I
2. A – IV, B – III, C – I, D – II
3. A – I, B – II, C – III, D – IV

26. A proton is projected with a speed $v$into a magnetic field $B$of magnitude 1 Twith the angle between the proton velocity and the magnetic field being $60°$, as shown. The kinetic energy of the proton is 2 eV(with the proton's mass $= 1.67 \times 10^{-27}$ kg, and charge $e = 1.6 \times 10^{-19}$ C). The pitch of the path of the proton is:



1. 6.28×10^{-2} m
2. 3.14×10^{-4} m
3. 6.42×10^{-4} m
4. 3.14×10^{-2} m

27. The magnifying power of a telescope is 9. When it is adjusted for parallel rays the distance between the objective and eyepiece is 20 cm. The focal length of the lenses is:
1. 10 cm, 10 cm
2. 15 cm, 5 cm

3. 3. 18 cm, 2 cm
4. 11 cm, 9 cm

28. A light ray falls on a glass surface of refractive index√3, at an angle of 60°. The angle between the refracted and reflected rays would be:

1.	120°	2.	30°
3.	60°	4.	90°

29. When a soft-iron core is inserted into a solenoid, themagnetic magnetic field is observed to increase by 1500 times near its opening. The turns density is 1000 m^{-1} and the solenoid carries a current 100 mA. The field at its centre is (nearly):
1. 0.2 T
2. 0.02 T
3. 0.4 T
4. 0.04 T

30. The ratio of contributions made by the electric field and magnetic field components to the intensity of an electromagnetic wave is: ($c =$ speed of electromagnetic waves)

1. $1:1$
2. $1:c$
3. $1:c^2$
4. $c:1$

31. A body moves with uniform acceleration. If its velocity changes from 10 m/sto 20 m/sin 5 s, the acceleration is:

1. 1 m/s^2
2. 2 m/s^2
3. 3 m/s^2
4. 4 m/s^2

32. The work done by a force is maximum when the angle between force and displacement is:

1. 0°
2. 30°
3. 60°
4. 90°

33. The dimensional formula of force is:

1. $[MLT^{-2}]$
2. $[ML^2T^{-2}]$
3. $[ML^{-1}T^{-2}]$
4. $[M^0LT^{-2}]$

34. The potential energy of a spring is given by:

1. $\frac{1}{2}kx$

2. $\frac{1}{2}kx^2$

3. kx^2

4. kx

35. If the frequency of a wave increases, its wavelength:

1. Increases

2. Decreases

3. Remains constant

4. Becomes zero

36. The electric field inside a conductor is:

1. Maximum

2. Minimum

3. Zero

4. Infinite

37. The SI unit of magnetic flux is:

1. Tesla

2. Weber

3. Henry

4. Ohm

38. The current in a conductor is due to the flow of:

1. Protons

2. Neutrons
3. Electrons
4. Photons

39. The refractive index of a medium depends on:

1. Frequency of light
2. Speed of light
3. Both frequency and medium
4. Only medium

40. The energy of an electron in the ground state of hydrogen atom is:

1. 13.6 eV
2. −13.6 eV
3. 0 eV
4. 27.2 eV

41. The time period of oscillation is inversely proportional to:

1. Frequency
2. Amplitude
3. Velocity
4. Displacement

42. In an isothermal process, which quantity remains constant?

1. Pressure
2. Volume
3. Temperature
4. Internal energy

43. The half-life of a radioactive substance depends on:

1. Initial quantity
2. Temperature
3. Pressure
4. Nature of substance

44. The capacitance of a capacitor increases when:

1. Distance between plates increases
2. Area of plates decreases
3. Dielectric constant increases
4. Voltage increases

45. The speed of light in vacuum is approximately:

1. 3×10^6 m/s
2. 3×10^7 m/s
3. 3×10^8 m/s

4. 3×10^9 m/s

CHEMISTRY SECTION

46. Which of the following solutions will cause Mohr's salt to precipitate out as iron (II) hydroxide?

1. Hydrochloric acid 2. Sodium hydroxide

3. Sulphuric acid

4. Potassium nitrat

47. Match List-I with List-II:

List-I (Atom/Molecule)
A. Nitrogen
B. Fluorine molecule
C. Oxygen molecule
D. Xenon atom

List-II (Property)
I. Paramagnetic
II. Most reactive element in group 18
III. Element with highest ionisation enthalpy in group 15
IV. Strongest oxidising agent

Identify the correct answer from the options given below:

1. A – III, B – I, C – IV, D – II
2. A – I, B – IV, C – III, D – II
3. A – II, B – IV, C – I, D – II
4. A – III, B – IV, C – I, D – II

48. What is the value of ΔH for the reaction $2Al + Cr_2O_3 \rightarrow 2Cr + Al_2O_3$, knowing that the enthalpies of

formation of Al_2O_3 and Cr_2O_3 are –1596 kJ and –1134 kJ, respectively?

1.–1365 kJ

3.–2730 kJ

2. 2730 kJ

4.–462 k

49. Match List-I with List-II

	List-I (Atom/Molecule)		List-II (Property)
A.	Nitrogen	I.	Paramagnetic
B.	Fluorine molecule	II.	Most reactive element in group 18
C.	Oxygen molecule	III.	Element with highest ionisation enthalpy in group 15
D.	Xenon atom	IV.	Strongest oxidising agent

Identify the correct answer from the option given below:

1.	A- III, B-I, C-IV, D-II	2.	A-I, B-IV, C-III, D-II
3.	A-II, B-IV, C-I, D-II	4.	A-III, B-IV, C-I, D-II

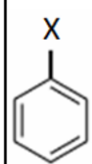
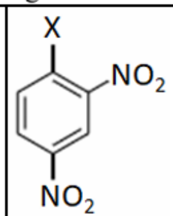
50. Assertion (A): SeCl4 does not have a tetrahedral structure.
Reason (R): Se has two lone pairs in SeCl4 Both (A) and (R) are True and (R) is the correct explanation of (A).

1. Both (A) and (R) are True but (R) is not the correct explanation of (A).

2. (A) is True but (R) is False.

3. Both (A) and (R) are False.

51. The correct order of increasing C–X bond reactivity toward nucleophiles among the following is:

I		**II**	
III	$(CH_3)_3C-X$	**IV**	$(CH_3)_2CH-X$

1.	I < II < IV < III	2.	II < III < I < IV
3.	IV < III < I < II	4.	III < II < I < IV

52. Given that

$$E^\circ_{O_2/H_2O} = +1.23 \text{ V}$$
$$E^\circ_{S_2O_8^{2-}/SO_4^{2-}} = 2.05 \text{ V}$$
$$E^\circ_{Br_2/Br^-} = +1.09 \text{ V}$$
$$E^\circ_{Au^{3+}/Au} = +1.4 \text{ V}$$

The strongest oxidising agent is:

1. Au^{3+}
2. O_2
3. $S_2O_8^{2-}$
4. Br_2

53. What is the oxidation state of a phosphorus atom in hypophosphoric acid?

1. $+3$

2. +4

3. +5

4. +3

54. The lowest enthalpy of atomization among the following is:

1. Sc

2. Cu

3. Zn

4. Ni

55. Which of the following statements is correct for an element that forms oxides in different oxidation states?

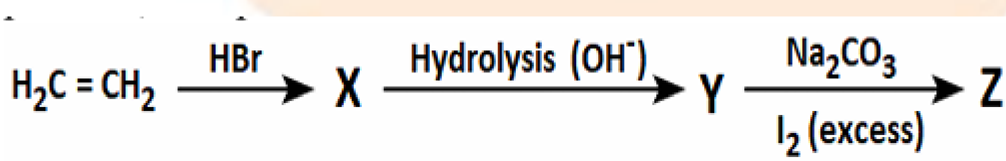
1. The oxide is neutral when the element is in its highest oxidation state.

2. The oxide is most acidic when the element is in its highest oxidation state.

3. The oxide is amphoteric when the element is in its highest oxidation state.

4. The oxide is most basic when the element is in its highest oxidation state.

56. In the following reaction sequence, what is the final product, compound Z?



1. C_2H_5I

2. CHI_3

3. CH_3CHO

4. C_2H_5OH

57. Match the species in Column-I with their corresponding hybridisation in Column-II:

	Column-I (Species)		Column-II (Hybridisation)
A.	Boron in $[B(OH)_4]^-$	i.	sp^2
B.	Aluminium in $[Al(H_2O)_6]^{3+}$	ii.	sp^3
C.	Carbon in Buckminsterfullerene	iii.	sp^3d^2
D.	Germanium in $[GeCl_6]^{2-}$		

Codes:

	A	B	C	D
1.	ii	iii	i	iii
2.	i	i	iii	ii
3.	iii	ii	ii	i
4.	i	iii	ii	iii

58. The compound among the following that used in cosmetic surgery is:

1. Silica 2. Silicates 3. Silicones 4. Zeolites

59. The enolic form of an acetone contains:

1. 9 sigma bonds, 1 pi bond, and 2 lone pairs of electrons

2. 8 sigma bonds, 2 pi bonds, and 2 lone pairs of electrons

3. 10 sigma bonds, 1 pi bond, and 1 lone pair of electrons

4. 9 sigma bonds, 2 pi bonds, and 1 lone pair of electrons

60. What factor influences the molar mass of acetic acid in benzene when measured using freezing point depression?

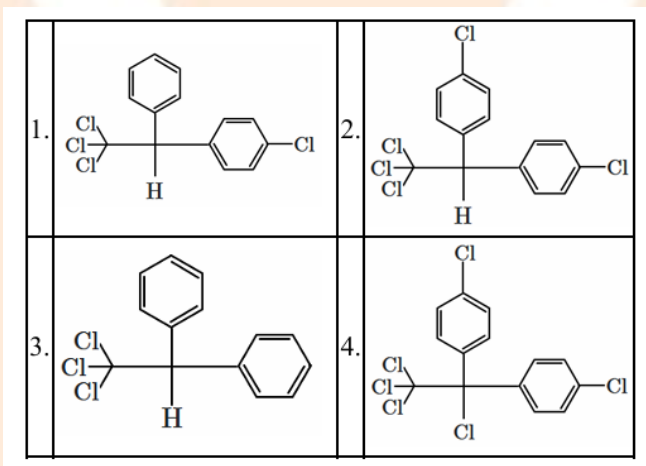
1. Vapour Pressure

2. Dissociation

3. Complex formation

4. Association (Dimerisation)

61. The correct structure of DDT is:



62. When two liquids A and B mix together, the boiling point of solution is found higher than that of the individual liquids. The nature of the solution is:

1.	Ideal solution.
2.	Positive deviation with a non-ideal solution.
3.	Negative deviation with a non-ideal solution.
4.	Normal solution.

63. Which structure(s) of proteins remains(s) intact during denaturation process?

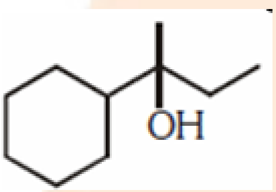
1. Both secondary and tertiary structures

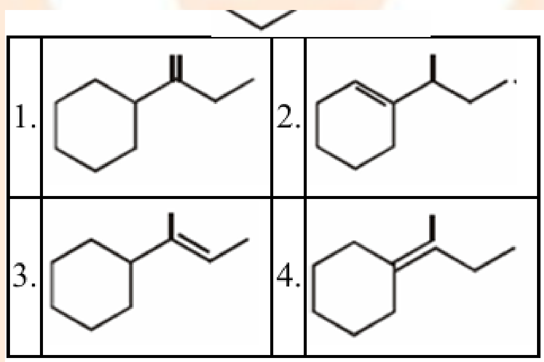
2. Primary structure only

3. Secondary structure only

4. Tertiary structure only

64. Which of the following is not the product of dehydration of -



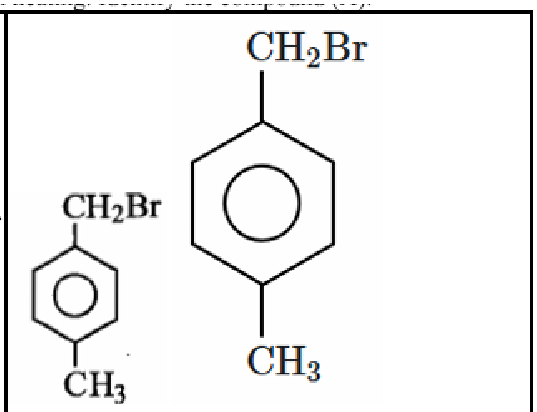


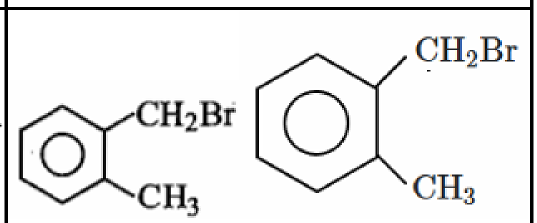
65. Which species in the given-below reversible reaction is/are acting as Brønsted acids?

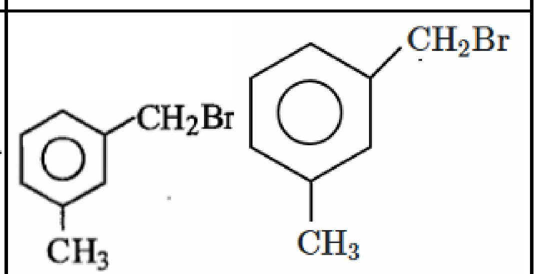
$$CO_3^{2-}(aq) + H_2O(l) \rightleftharpoons HCO_3^{-}(aq) + OH^{-}(aq)$$

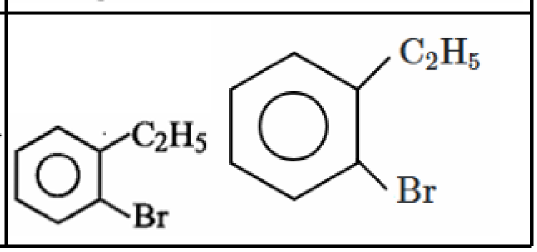
1.	$CO_3^{2-}(aq)$
2.	$H_2O(l)$ and $OH^-(aq)$
3.	$H_2O(l)$ and $HCO_3^-(aq)$
4.	$CO_3^{2-}(aq)$ and $OH^-(aq)$

66. Compound (A), C8H9Br , gives a white precipitate when warmed with alcholic AgNO3. Oxidation of (A) gives an acid (B), C8H6O4. (B) easily forms anhydride on heating. Identify the compound (A)

1. 

2. 

3. 

4. 

67. The correct value of cell potential in volts for the reaction that occurs when the following two half cells are connected, is:

$$Fe^{2+}_{(aq)} + 2e^- \rightarrow Fe(s), E^\circ = -0.44\ V$$

$$Cr_2O_7^{2-}{}_{(aq)} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O,$$

$$E^\circ = +1.33\ V$$

1. +1.77 V

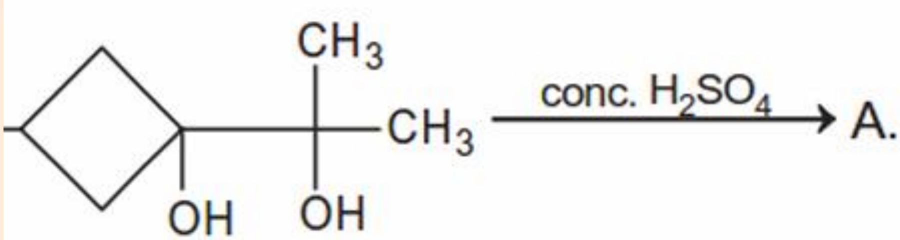
2. +2.65 V

3. +0.01 V

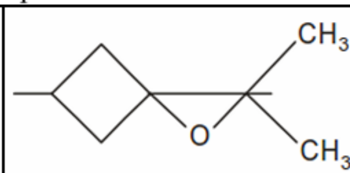
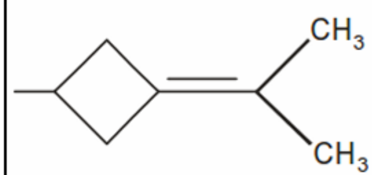
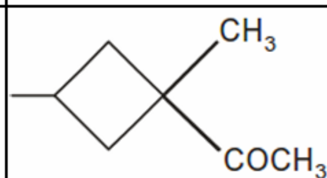
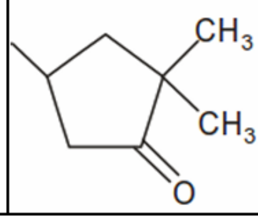
4. +0.89 V

68. In the reaction:

. In the reaction



The product is –

1.	
2.	
3.	
4.	

69. The number of moles of KMnO4 required to oxidize one mole of KI in the presence of sulphuric acid are:

1.	5	2.	2
3.	1/2	4.	1/5

70. Carbon and oxygen combine to form two oxides, carbon monoxide and carbon dioxide in which the ratio of the weights of carbon and oxygen is respectively 12 :16 and 12 : 32. These figures illustrate the:

1. Law of multiple proportions

2. Law of reciprocal proportions

3. Law of conservation of mass

4. Law of constant proportions

71. The number of moles of oxygen atoms in 44 g of CO_2 is:

1. 1
2. 2
3. 3
4. 4

72. Which of the following has the highest electronegativity?

1. F
2. Cl
3. O
4. N

73. The hybridization of carbon in CH_4 is:

1. sp
2. sp^2
3. sp^3
4. dsp^2

74. The bond order of O_2 is:

1. 1

2. 2

3. 3

4. 0

75. Which of the following is a strong electrolyte?

1. CH_3COOH

2. NH_4OH

3. HCl

4. H_2O

76. The pH of a neutral solution at 25°C is:

1. 0

2. 7

3. 14

4. 1

77. Which of the following is an example of a redox reaction?

1. $NaCl \rightarrow Na^+ + Cl^-$

2. $HCl + NaOH \rightarrow NaCl + H_2O$

3. $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$

4. $AgNO_3 + NaCl \rightarrow AgCl + NaNO_3$

78. Which of the following gases deviates most from ideal behavior?

1. H_2
2. He
3. NH_3
4. Ne

79. The oxidation state of sulphur in H_2SO_4 is:

1. +2
2. +4
3. +6
4. 0

80. Which of the following is a transition element?

1. Na
2. Mg
3. Fe
4. Al

81. Which of the following is the correct order of atomic radius?

1. Na > Mg > Al
2. Mg > Na > Al
3. Al > Mg > Na

4. Na > Al > Mg

82. Which of the following is most acidic?

1. CH_4
2. NH_3
3. H_2O
4. HF

83. The rate of a reaction doubles when temperature increases by 10°C. This is explained by:

1. Boyle's law
2. Arrhenius equation
3. Dalton's law
4. Henry's law

84. Which of the following is a noble gas?

1. N_2
2. O_2
3. Ne
4. CO_2

85. Which of the following is an example of a coordination compound?

1. NaCl
2. NH_3

3. $[Cu(NH_3)_4]^{2+}$
4. HCl

86. The IUPAC name of CH_3COOH is:

1. Methanoic acid
2. Ethanoic acid
3. Propanoic acid
4. Butanoic acid

87. Which of the following shows hydrogen bonding?

1. CH_4
2. NH_3
3. CO_2
4. HCl

88. Which of the following is an aldehyde?

1. CH_3COOH
2. CH_3CHO
3. CH_3OH
4. CH_3COCH_3

89. Which of the following compounds gives iodoform test?

1. CH_3CHO

2. HCHO

3. HCOOH

4. CH_3OH

90. Which of the following is a polymer?

1. Glucose

2. Ethene

3. PVC

4. Methane

BIOLOGY

91. In a dorsiventral leaf, palisade mesophyll cells are:

A) Loosely arranged for gaseous exchange
B) Compactly arranged for maximum light absorption
C) Located below lower epidermis
D) Lacking chloroplasts

92. Which feature is diagnostic of monocot stem T.S.?

A) Vascular bundles in a ring
B) Presence of cambium
C) Scattered vascular bundles with closed type
D) Presence of pith

93. Assertion (A): Collenchyma cells are living at maturity.
Reason (R): Their cell walls are uniformly lignified.

A) Both true, R explains A
B) Both true, R incorrect explanation
C) A true, R false
D) Both false

94. Photorespiration primarily occurs in:

A) Bundle sheath cells
B) Guard cells
C) Mesophyll cells
D) Epidermal cells

95. In C_4 plants, initial CO_2 fixation occurs in:

A) Mesophyll cells by PEP carboxylase
B) Bundle sheath cells by Rubisco
C) Guard cells by RuBP
D) Phloem cells

96. Which condition enhances photorespiration?

A) High CO_2
B) Low O_2
C) High temperature
D) Low light intensity

97. Water potential becomes more negative due to:

A) Increase in pressure potential
B) Increase in solute concentration
C) Decrease in solute concentration
D) Cell becoming turgid

98. Microtubules are composed of:

A) Actin
B) Tubulin
C) Keratin
D) Myosin

99. Which phase of mitosis ensures equal chromosome distribution?

A) Prophase
B) Metaphase
C) Anaphase
D) Telophase

100. Assertion (A): Mitochondria are semi-autonomous.
Reason (R): They possess their own circular DNA and ribosomes.

A) Both true, R explains A
B) Both true, R incorrect explanation
C) A true, R false
D) Both false

101. In DNA replication, leading strand is synthesized:

A) Continuously in 5'→3' direction
B) Discontinuously
C) In 3'→5' direction
D) Without primer

102. Which enzyme removes RNA primers?

A) Ligase
B) DNA polymerase I
C) Helicase
D) Topoisomerase

103. The wobble hypothesis explains:

A) DNA replication errors
B) Flexibility of third codon base pairing
C) Mutation rate
D) Gene regulation

104. Incomplete dominance differs from codominance because:

A) Only one allele expresses
B) Both alleles express equally
C) Phenotype is intermediate
D) No heterozygote formed

105. Linkage reduces:

A) Recombination frequency
B) Mutation rate
C) Allele frequency
D) Dominance

106. Assertion (A): Dominance is a relationship between alleles.
Reason (R): It depends on gene product interaction.

A) Both true, R explains A
B) Both true, R incorrect
C) A true, R false
D) Both false

107. Hardy–Weinberg equilibrium is disturbed by:

A) Random mating
B) Large population
C) Mutation
D) Absence of selection

108. Adaptive radiation is best explained by:

A) Industrial melanism
B) Darwin's finches
C) Lamarckism
D) Genetic drift

109. During ventricular diastole:

A) Semilunar valves open
B) AV valves open
C) Both closed
D) Both open

110. The pacemaker of heart is:

A) AV node
B) SA node
C) Purkinje fibers
D) Bundle of His

111. ADH primarily acts on:

A) Loop of Henle
B) Collecting duct
C) PCT
D) DCT only

112. Bohr effect refers to:

A) O_2 binding to hemoglobin
B) CO_2 affecting O_2 affinity
C) pH increase
D) Carbonic acid formation

113. Depolarisation involves:

A) Na^+ efflux
B) K^+ influx
C) Na^+ influx
D) Cl^- influx

114. Myelin sheath increases:

A) Membrane permeability
B) Conduction velocity
C) Synaptic delay
D) Action potential amplitude

115. Clonal selection theory explains:

A) Antibody diversity
B) Antigen mutation
C) Viral replication
D) Phagocytosis

116. Primary immune response is characterized by:

A) Rapid high IgG
B) Delayed IgM
C) No memory cells
D) IgE dominance

117. Ecological pyramids are always upright for:

A) Energy
B) Biomass
C) Numbers
D) Producers

118. Gause's principle refers to:

A) Competitive exclusion
B) Food chain
C) Ecological succession
D) Carrying capacity

119. Standing crop differs from standing state in terms of:

A) Units
B) Biomass
C) Productivity
D) Trophic level

120. Double fertilisation results in:

A) Zygote + Endosperm
B) Two zygotes
C) Embryo only
D) Endosperm only

121. Tapetum is important for:

A) Pollen wall formation
B) Fertilisation
C) Seed dispersal
D) Embryo growth

122. LH surge causes:

A) Follicle growth
B) Ovulation
C) Endometrial shedding
D) Progesterone decline

123. Progesterone maintains:

A) Follicular phase
B) Secretory phase
C) Menstrual flow
D) Ovulation

124. Restriction enzymes recognize:

A) Random sequences
B) Palindromic sequences
C) Promoters
D) Introns

125. PCR requires:

A) Reverse transcriptase
B) DNA polymerase
C) Ligase
D) RNA primer

126. Ozone depletion mainly due to:

A) CO_2
B) CFCs
C) NO_2
D) SO_2

127. Eutrophication results from:

A) Heavy metals
B) Excess nutrients
C) Acid rain
D) Deforestation

128. Which of the following correctly describes the Meselson–Stahl experiment outcome after two generations in ^{14}N medium?

A) All heavy DNA
B) All light DNA
C) 50% hybrid and 50% light DNA
D) 25% heavy and 75% hybrid

129. In lac operon, the repressor binds to:

A) Promoter
B) Operator
C) Structural gene
D) CAP site

130. The anticodon loop is found in:

A) mRNA
B) rRNA
C) tRNA
D) snRNA

131. Assertion (A): DNA replication is semi-discontinuous.
Reason (R): Leading strand is synthesized continuously and lagging strand discontinuously.

A) Both true, R explains A
B) Both true, R incorrect
C) A true, R false
D) Both false

132. In a dihybrid cross, phenotypic ratio 9:3:3:1 appears only when:

A) Genes are linked
B) Genes assort independently
C) Complete dominance absent
D) Epistasis occurs

133. Mutation in germ cells leads to:

A) Somatic disorders
B) Inherited variation
C) Cancer only
D) Immediate lethality

134. Genetic drift is more significant in:

A) Large populations
B) Small populations
C) Mutating populations
D) Stable populations

135. Which is an example of convergent evolution?

A) Homologous wings
B) Analogous wings
C) Vertebrate forelimbs
D) Pentadactyl limb

136. Founder effect results from:

A) Natural selection
B) Genetic drift
C) Mutation pressure
D) Recombination

137. During cyclic photophosphorylation:

A) NADPH formed
B) O_2 released
C) Only ATP formed
D) Both ATP and NADPH formed

138. C_4 pathway advantage under high temperature is due to:

A) Reduced transpiration
B) High Rubisco activity
C) Suppression of photorespiration
D) Increased stomatal density

139. Which hormone promotes bolting in rosette plants?

A) Auxin
B) Cytokinin
C) Gibberellin
D) Ethylene

140. During inspiration:

A) Diaphragm relaxes
B) Thoracic volume decreases
C) Intrapulmonary pressure increases
D) Diaphragm contracts

141. Counter-current mechanism operates in:

A) PCT
B) Loop of Henle
C) DCT
D) Bowman's capsule

142. Bohr effect enhances oxygen release at:

A) High pH
B) Low CO_2
C) High CO_2 concentration
D) Low temperature

143. Which ion primarily triggers neurotransmitter release?

A) Na^+
B) K^+
C) Ca^{2+}
D) Cl^-

144. Saltatory conduction occurs due to:

A) Continuous depolarization
B) Nodes of Ranvier
C) Lack of myelin
D) Increased axon diameter only

145. Repolarization involves:

A) Na^+ influx
B) K^+ efflux
C) Ca^{2+} influx
D) Cl^- influx

146. Secondary immune response is:

A) Slow, low antibody titer
B) Rapid, high IgG
C) IgM dominant
D) Without memory cells

147. MHC molecules are essential for:

A) Antibody production
B) Antigen presentation
C) Complement activation
D) Phagocytosis

148. Standing state is measured in:

A) Energy units
B) Biomass
C) Nutrient amount
D) Trophic levels

149. Primary succession starts on:

A) Forest
B) Pond
C) Bare rock
D) Grassland

150. GPP − R =

A) NPP
B) Biomass
C) Standing crop
D) Productivity

151. Triple fusion produces:

A) Diploid zygote
B) Triploid endosperm
C) Haploid embryo
D) Polyploid nucellus

152. Self-incompatibility prevents:

A) Self-fertilization
B) Pollination
C) Seed formation
D) Cross pollination

153. Progesterone level peaks during:

A) Follicular phase
B) Ovulation
C) Luteal phase
D) Menstrual phase

154. Spermatogenesis completes in:

A) Epididymis
B) Seminiferous tubules
C) Prostate
D) Vas deferens

155. DNA ligase joins:

A) Hydrogen bonds
B) Phosphodiester bonds
C) Glycosidic bonds
D) Peptide bonds

156. Gene cloning requires:

A) Vector
B) Restriction enzyme
C) DNA ligase
D) All of these

157. Hotspots are identified based on:

A) Species richness only
B) Endemism and threat level
C) Area
D) Population density

158. Biological magnification increases across:

A) Producers
B) Trophic levels
C) Habitat
D) Seasons

159. Assertion: Crossing over increases genetic variation.
Reason: It occurs between non-sister chromatids.

A) Both true, R explains
B) Both true, R not explanation
C) A true, R false
D) Both false

160. Assertion: Transpiration pull is cohesive.
Reason: Water molecules exhibit hydrogen bonding.

A) Both true, R explains
B) Both true, R not explanation
C) A true, R false
D) Both false

161. Mutation in HBB gene leads to:

A) Hemophilia
B) Sickle-cell anemia
C) Thalassemia only
D) Color blindness

162. High CO_2 in blood shifts oxyhemoglobin dissociation curve:

A) Left
B) Right
C) No shift
D) Downward

163. Cyclic AMP functions as:

A) Hormone
B) Second messenger
C) Enzyme
D) Neurotransmitter

164. Phytochrome exists in:

A) Single form
B) Two interconvertible forms
C) Three forms
D) Inactive form only

165. Which organism performs chemosynthesis?

A) Cyanobacteria
B) Nitrosomonas
C) Algae
D) Fungi

166. Which enzyme unwinds DNA?

A) Ligase
B) Helicase
C) Primase
D) Endonuclease

167. Adaptive radiation in mammals occurred after:

A) Cambrian explosion
B) Dinosaur extinction
C) Ice age
D) First amphibians

168. Transcription in eukaryotes occurs in:

A) Cytoplasm
B) Ribosome
C) Nucleus
D) Golgi

169. Most abundant protein in blood plasma:

A) Globulin
B) Fibrinogen
C) Albumin
D) Hemoglobin

170. The trophic level with maximum energy loss:

A) Producers
B) Primary consumers
C) Secondary consumers
D) Between each transfer

171. Cytokinesis in plant cells occurs by:

A) Cleavage furrow
B) Cell plate formation
C) Budding
D) Binary fission

172. Restriction enzymes are obtained from:

A) Plants
B) Animals
C) Bacteria
D) Viruses

173. The main cause of greenhouse effect:

A) Ozone
B) CO_2
C) O_2
D) N_2

174. Which process ensures genetic stability?

A) Mutation
B) DNA repair
C) Crossing over
D) Recombination

175. Which layer forms pollen wall exine?

A) Tapetum
B) Intine
C) Epidermis
D) Microspore

176. Dominant allele masks recessive due to:

A) Allele frequency
B) Protein functionality
C) Mutation
D) Linkage

177. Insulin lowers blood glucose by:

A) Glycogenolysis
B) Gluconeogenesis
C) Glycogenesis
D) Lipolysis

178. Ecosystem stability increases with:

A) Low diversity
B) High diversity
C) Single trophic level
D) High disturbance

179. Which is an in situ conservation method?

A) Botanical garden
B) Gene bank
C) National park
D) Tissue culture

180. Photosynthesis redox nature demonstrated by:

A) Hill
B) Van Niel
C) Arnon
D) Calvin